

CrumbBob gives IBM Bob a durable memory layer.

It turns one-time Bob session reports into CRUMB v1.4 packs, proof chains, replay prompts, and queryable team memory.

290

tests passed

65.17%

verified coverage

9

pack artifacts

SHA-256

proof chain

Capture. Compress. Replay. Prove.

Bob understands the repo during the session. Then the context disappears.

The handoff problem is not markdown conversion. It is preserving enough structured evidence for the next agent, reviewer, or teammate to continue without rediscovery.



CrumbBob replaces this weak handoff with typed CRUMB files, hashes, extracted counts, and replayable next-session instructions.

One report becomes a verifiable intelligence system.

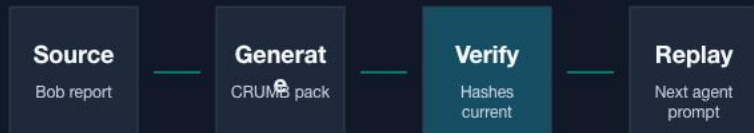
The product path is intentionally local-first: files, SQLite, CLI, web dashboard, and optional LLM analysis.



Design principle: keep the core useful without cloud services, then add LLM and web experiences as opt-in layers.

Judge credibility comes from hashes, counts, and replayable artifacts.

Source input, generated output, extracted counts, and replay instructions are tied together.



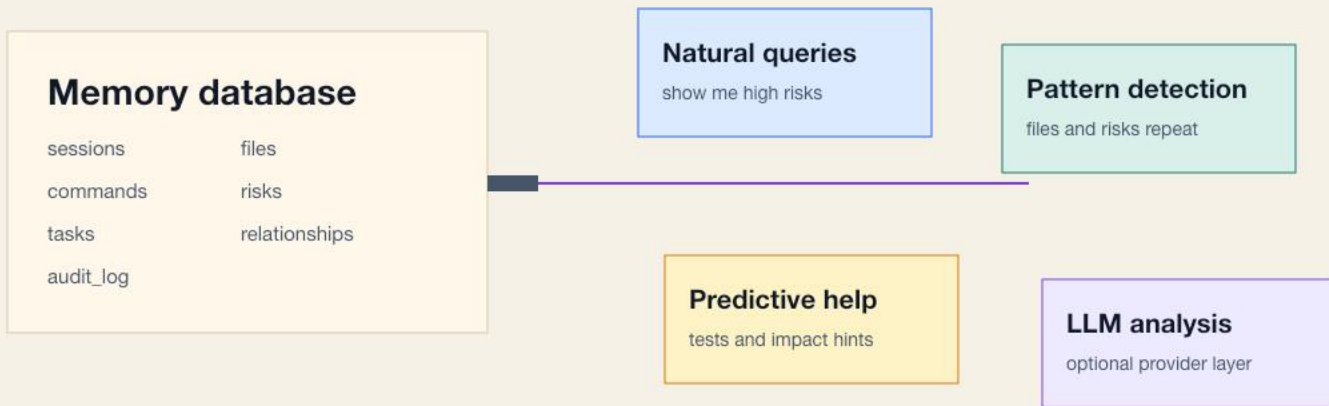
08_proof_chain.json

source_report.sha256	8f4e...d91a
generated_files[9]	sha256 per artifact
extracted_counts	files, commands, risks,
tasks	
crumbob_version	0.3.1
replay_prompt	present
pr_summary	present

Doctor verdict: ready for judge walkthrough

Session memory turns isolated Bob reports into team intelligence.

SQLite stores sessions, files, commands, risks, tasks, relationships, insights, and audit events so later queries can use history.



The judge walkthrough is five commands, not a custom setup.

The demo uses checked-in example data and produces visible trust signals: validate, doctor, replay, query, and dashboard.

```
$ crumbob pack examples/compliance-ai --out /tmp/demo
$ crumbob doctor /tmp/demo
verdict: ready for judge walkthrough

$ crumbob init-db
$ crumbob record /tmp/demo
$ crumbob query natural "show me high risks"

$ crumbob serve --no-browser
open http://localhost:8000
```

What judges see

- Valid CRUMB files
- Current proof hashes
- Replay prompt
- PR summary
- Dashboard data

v0.3.1 is submission-ready because the gate is measurable.

The final pass verified tests, typing, linting, package build, dependency audit, security scan, and example pack health.

Tests

290 passed

Coverage

65.17%

Types

mypy clean

Lint

Ruff clean

Security

Bandit clean

Deps

pip-audit clean

Package

build + twine

Demo

doctor ready

CrumbBob is the missing memory layer around IBM Bob

It is useful as a hackathon demo today and credible as an enterprise workflow tomorrow: local-first, verifiable, tested, and easy to inspect.

Final handoff

- README tells the product story
- SUBMISSION.md frames the judging case
- CODEX_SUBMISSION_PROMPT.md is ready to reuse
- Proof Chain shows replayable evidence
- Demo commands run locally

Capture. Compress. Replay. Prove.